

Onboarding Drop-off Analyzer for Growth Teams

Hackathon Submission

1. Executive Summary

User onboarding is one of the most critical stages of the growth funnel. However, most teams struggle to pinpoint exactly where and why users drop off. Our AI-powered Onboarding Drop-off Analyzer integrates with product analytics tools (PostHog, Mixpanel, Google Sheets) and provides weekly insights into friction points, error clusters, and device-specific bottlenecks. It further recommends actionable UX improvements, automatically delivered to Slack and stored for long-term tracking.

- Identify drop-off points in onboarding.
- Cluster reasons behind drop-offs.
- Recommend 2–3 actionable UX fixes every week.
Reports are auto-delivered to Slack and stored in Google Sheets, making growth teams proactive instead of reactive.

2. Problem & Solution

Problem Statement:

- Analytics tools show *where* users drop off but not *why*.
- Growth teams waste cycles guessing fixes.
- Result: poor retention, high churn, wasted marketing spend.

Solution Overview:

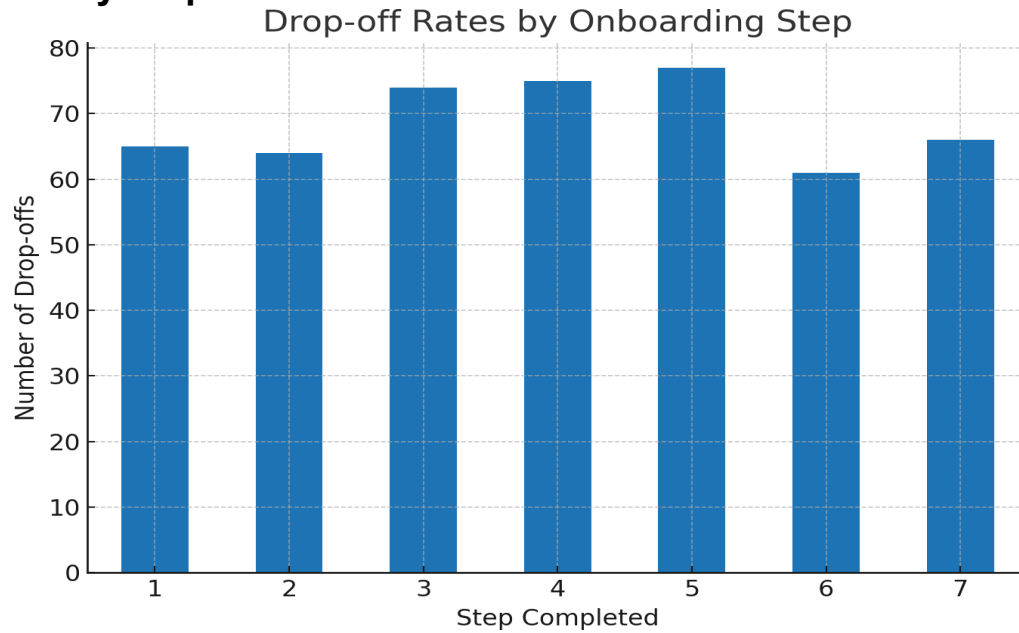
The Onboarding Drop-off Analyzer solves this by:

- Aggregating onboarding events.
- Clustering drop-off causes (UX friction, device issues, validation errors).
- Using Google Gemini AI to recommend targeted fixes.
- Delivering automated reports weekly to Slack & Sheets.

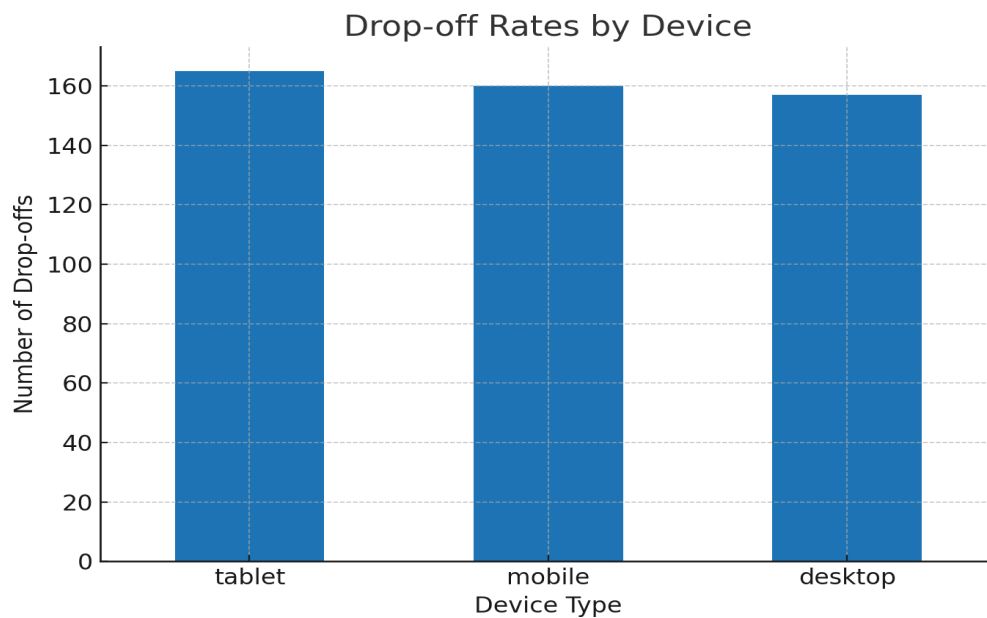
3. Sample Insights from Mock Data

Below are sample insights generated from our mock onboarding dataset

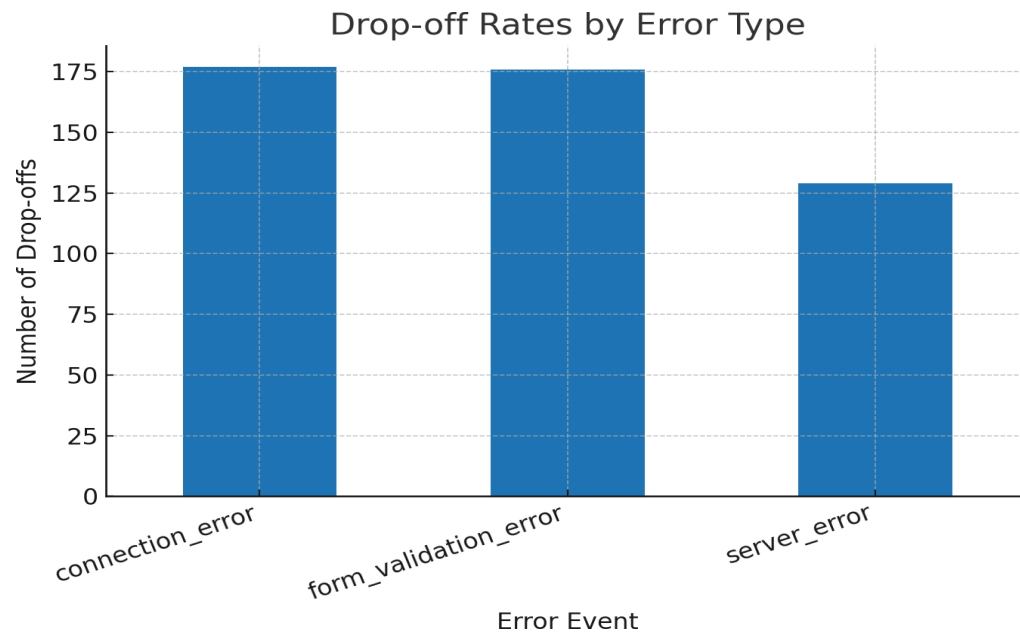
Drop-off by Step



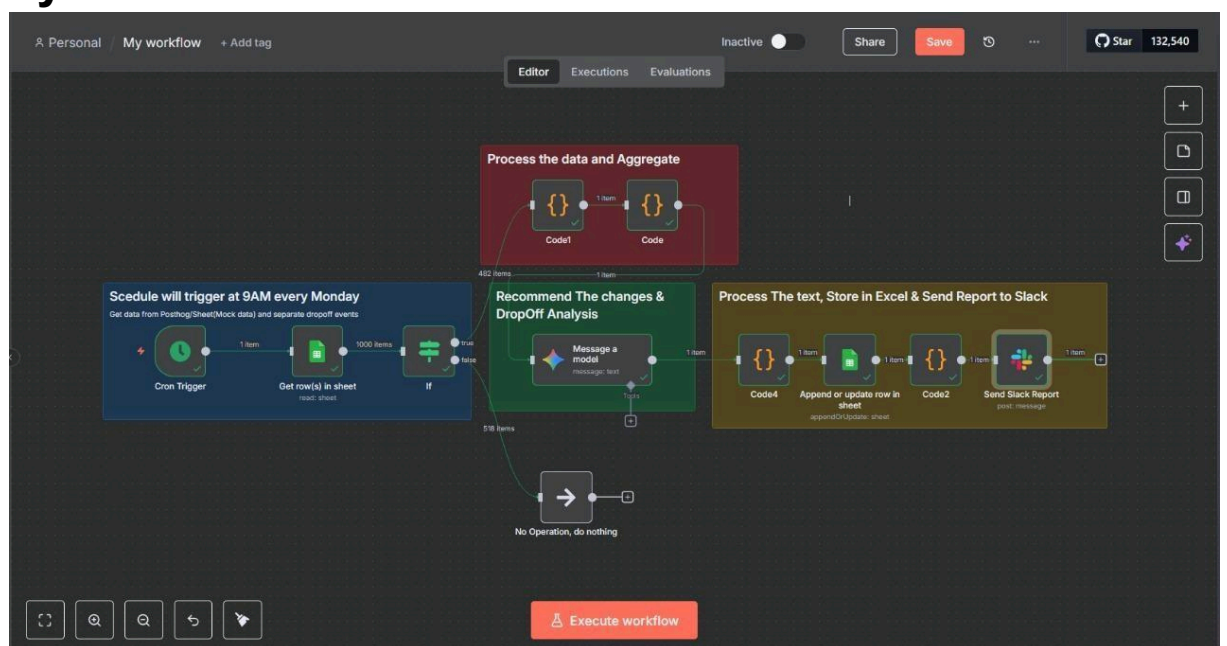
Drop-off by Device



Drop-off by Error Event



4. System Architecture



The system is built on an automated workflow (implemented in n8n) that runs weekly, pulling onboarding data from Google Sheets/PostHog, filtering drop-offs, aggregating statistics, enriching them with contextual analysis, and passing them through Google Gemini for clustering and recommendations. Results are stored in Sheets and sent to Slack automatically.

5. Workflow Walkthrough

Step	Description
1. Trigger	Cron runs weekly at 9 AM.
2. Data Fetch	Retrieve onboarding event logs from Google Sheets/PostHog.
3. Drop-off Filter	Filter rows where drop_off = true.
4. Aggregation	Summarize drop-offs by step, device, error, and duration.
5. AI Analysis	Gemini clusters reasons and suggests fixes.
6. Report Storage	Append results to Google Sheets for history.
7. Report Delivery	Send insights automatically to Slack channels
8. Final Processing	Code node enriches/cleans the report before delivery.
9. Report Delivery	Slack node sends the onboarding drop-off analysis directly to team channels.
10. No Operation (optional)	If no drop-off events are found, workflow exits without action.

6. Impact & Future Enhancements

Impact on Growth Teams:

- Faster iteration cycles (weekly actionable fixes).
- Reduced onboarding friction.
- Improved activation & retention rates.

Future Enhancements:

- Real-time monitoring.
- Predictive analytics for churn prevention.
- Automated A/B testing for UX fixes.
- Multi-dimensional clustering (step + device + error).

7. Conclusion

Our AI-powered Onboarding Drop-off Analyzer transforms raw analytics into actionable growth insights. By combining automation, clustering, and AI-driven recommendations, it empowers teams to iterate faster, reduce friction, and boost user activation and retention rates. This makes it an invaluable tool for modern growth teams in competitive markets.